

1BY0 SOLUTIONS · LINESIGHT OEE

OEE Project Management Brief

Digital production monitoring for Nakshatra water bottle filling — combining OEE, TPM, 4M, Right First Time (RFT) and Kaizen into one management improvement system.

Plant: Nakshatra

Line-01 · Filling

90 BPM · 24 btl/case

Target 225 cases/hour

1. Executive Summary & Why OEE

One digital system to monitor production, downtime, quality and OEE — then drive loss reduction.

OBJECTIVE

Capture hourly production + downtime, calculate OEE automatically, and give management loss visibility by hour, machine, SKU, reason and shift.

PROBLEM TODAY

Data exists in Excel, but losses are hard to analyse quickly. Manual reporting delays visibility and introduces formula / entry errors.

OUTCOME

Faster decisions, clearer accountability, reduced chronic losses and better line utilisation — with one KPI language across shifts.

OEE = Availability × Performance × Quality

Availability
OPERATING ÷ PLANNED TIME

Was the line running when it should? Losses: breakdowns, utilities, manpower, material, planned stops.

Performance
ACTUAL VS IDEAL OUTPUT

Did it run at expected speed? Losses: speed loss, minor stops, adjustments, process variation.

Quality
GOOD ÷ TOTAL OUTPUT

How much good product? Losses: fill, cap, bottle, label, coding, packaging defects.

Line-01 reference (theoretical target)

90

BPM filler

24

Bottles / case

225

Cases / hour

08–18

Shift window

90 BPM × 60 = 5,400 bottles/hour ÷ 24 = **225 cases/hour**. Lunch and approved planned stops must follow the agreed OEE policy so the denominator is consistent for management.

Management message: Move from reporting numbers to *managing losses*. OEE tells us where we lose; improvement systems (TPM / Kaizen / RFT) remove why we lose.

2. Digital OEE System — Scope & Operating Loop

From shop floor capture to management action.

Project scope

Masters & Plan

Plant, Line, Machine, SKU, Pack Size, Rated Speed, Shift, Users · Hourly target, SKU, planned quantity, schedule.

Shop floor capture

Hourly target / actual / loss / running time · Downtime start-end, category, reason, machine · Quality good / reject / rework.

Analysis

Shift / day / line / machine / loss dashboards · Pareto of downtime minutes · Target vs Actual vs Loss.

Outputs

Excel / PDF reports · Role-based access (Operator → Supervisor → Admin / Management) · Drill-down Plant → Line → Machine → SKU → Hour → Reason.

Closed-loop data flow

1. Plan

Hourly target & SKU

2. Capture

Production & downtime

3. Validate

Time, qty, reason

4. Calculate

$A \times P \times Q$

5. Analyze

Loss & trend

6. Act

Corrective action

Hourly visibility (example)

Time	Target	Actual	Loss	Typical reason
08–09	225	59	166	Startup
09–10	225	100	125	Breakdown
10–11	225	170	55	Changeover
11–12	225	200	25	—

Management can see immediately where target was missed and drill into the reason — without waiting for end-of-day Excel.

Ownership

Operator

Enter production, downtime, quality

Supervisor

Validate, review losses, shift actions

Management

Masters, KPI review, improvement decisions

3. TPM & 4M — From Losses to Projects

OEE identifies losses; TPM converts them into structured improvement projects.

OEE + TPM: OEE = KPI to measure effectiveness. TPM = system to eliminate equipment and process losses (zero breakdowns, zero defects, zero minor stops, operator ownership).

4M root-cause classification

MAN
People
 Operator skill, training, manpower, handling variation

MACHINE
Equipment
 Filler, capper, conveyor, electrical, mechanical, sensors

MATERIAL
Inputs
 Preform, bottle, cap, label, water, packaging quality

METHOD
Process
 SOP, settings, changeover, inspection, startup sequence

Classify every recurring loss → verify with 5-Why / evidence → remove the root cause (not only the symptom).

Proposed TPM projects (Line-01)

#	Project	OEE link	Focus
1	Filler breakdown reduction	Availability	Gearbox, motor, sensor, mechanical
2	Bottle damage reduction	Quality + Performance	Conveyor / guide / handling
3	Cap defect reduction	Quality	Capper settings & inspection
4	Startup loss reduction	Availability	Startup checklist & presets
5	Changeover time reduction	Availability	SMED + checklist
6	Minor stop elimination	Performance	Pareto short stops + PM

Improvement cycle

- Measure OEE

Pareto

4M + 5 Why

TPM / Kaizen

Standardize

Verify OEE

Recommended: every month pick top 3 losses and launch focused TPM projects. Track Before OEE → Countermeasure → After OEE → Sustenance.

4. Right First Time (RFT) — Quality that protects OEE

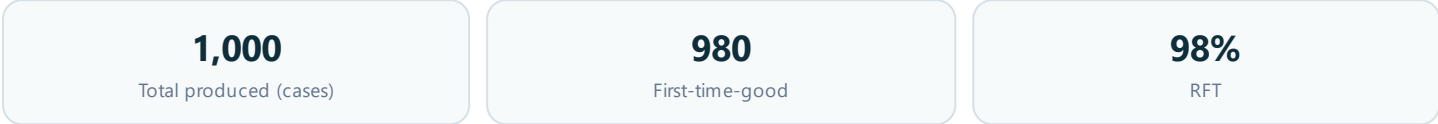
Make product correct the first time — without rework, repair or repeat processing.

RFT
 First-time quality performance

OEE link
 Improves the Quality pillar

Focus
 Prevention, not detection after the fact

$$\text{RFT \%} = \text{First-Time-Good Quantity} \div \text{Total Produced} \times 100$$



If 20 cases need rework before acceptance, they are *not* Right First Time. Define plant rules for rework / reject before using RFT as a KPI.

How OEE + RFT + TPM work together

OEE
 Measures overall equipment effectiveness and major losses

RFT
 Measures whether product was made correctly first time

TPM / 4M
 Eliminates root causes of defects and stops

RFT projects for water filling

Area	Goal	Levers
Filling accuracy	Reduce under/overflow	Valve, calibration, settings
Cap quality	Loose / damaged caps	Torque, alignment, cap supply
Bottle quality	Damage / deformation	Conveyor, guides, material
Label & coding	Correct label / date	Changeover check + first-piece OK
Packaging	Pack / shrink defects	Count, inspection, carton

Combined cycle for management: **OEE Pareto** → **RFT defect** → **4M analysis** → **countermeasure** → **verify RFT & OEE**

5. Manufacturing Kaizen — Continuous Improvement

Small, practical, daily improvements with owners, targets and before/after KPIs.

KAIZEN
 Continuous improvement

FOCUS
 Small changes, every day

PEOPLE
 Operator + Supervisor + Eng.

RESULT
 Safety · Quality · Delivery · Cost

PDCA method

PLAN
 Problem, target, root cause

DO
 Implement countermeasure

CHECK
 Before vs after KPI

ACT
 Standardize & sustain

Priority Kaizen slate (examples)

Kaizen	From → To	KPI
Startup loss	60 → 30 min	Availability
SKU changeover	40 → 20 min	Availability
Minor stops	120 → 60 min	Performance
Bottle damage	3% → <1%	RFT / Quality
Material waste	Baseline → target	Cost / Quality
Stable filler speed	Actual → standard	Performance

Project tracking format (use monthly)

Project	Before	4M	Action	After target	KPI
Bottle damage	3.0%	Machine	Guide alignment + PM	<1%	RFT
Startup loss	60 min	Method	Checklist + presets	<30 min	Availability
Minor stops	120 min	Machine	Sensor PM / standard	<60 min	Performance
Cap defects	2.0%	Material	Cap spec + control	<0.5%	RFT / Quality

Governance cadence

Daily
 Top loss + immediate action

Weekly
 Open projects + owners

Monthly
 Before / after KPI close

Quarterly
 Standardize & replicate

6. Roadmap, Benefits & Decisions Required

What management needs to approve to make Line-01 pilot successful.

Implementation roadmap

PHASE 1–2

Masters + Plan + Hourly entry

OEE definitions, production plan, hourly capture

PHASE 3–4

Downtime + Auto OEE

Reason master, calculation & validation

PHASE 5–6

Dashboard + Pilot

Reports, roles · Line-01 pilot → standardize → expand

Expected business benefits

- Real-time visibility of production and downtime
- Less manual Excel work and fewer calculation errors
- Faster identification of chronic losses
- Stronger accountability for downtime reasons
- Better planning and target achievement
- History for MTBF, MTTR, Pareto and continuous improvement
- One common KPI language across shifts and lines

Decisions required from management

1. OEE policy

Planned Production Time; how to treat startup, changeover, lunch

2. Standard masters

Downtime, quality and production-loss reason lists

3. Data ownership

Operator / Supervisor / Admin responsibilities

4. Review rhythm

Hourly ops review + shift/day management KPI review

5. Pilot approval

Confirm Line-01 pilot scope and success criteria

6. Rollout plan

When to expand to other lines after standardisation

Integrated framework: OEE (measure) → Pareto (select) → 4M + 5 Why (analyse) → Kaizen (improve) → TPM (prevent) → RFT (confirm quality) → Standardize (sustain).

7. Final Management Message

The objective is not only to calculate OEE — it is to use OEE to improve the line.

Measure → Identify → Analyze → Improve → Standardize → Sustain

What each system does

- **OEE**
— measures performance
- **RFT**
— protects first-time quality
- **4M + 5 Why**
— finds root causes
- **TPM**
— prevents recurrence
- **Kaizen**
— creates continuous improvement

Ask of management

- Approve Line-01 digital OEE pilot
- Lock OEE / RFT definitions and masters
- Assign owners for top 3 monthly losses
- Review Before / After KPIs every month
- Replicate wins across lines and SKUs

THANK YOU

Sustainable improvement — not only a better OEE number

Plant: Nakshatra · Line-01 Water Bottle Filling · LineSight by 1BY0 Solutions